

A stabilized nonconforming finite element method for the surface biharmonic problem

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Outline

- 1 Introduction
- 2 Surface New Zienkiewiz-Type Element
- 3 Stabilized Nonconforming FEM
- 4 Numerical Experiments

Geometric PDEs



(a) Thin shells



(b) Biomembranes



(c) Surface Stokes

Three classes of finite element method for the PDEs posed on surfaces:

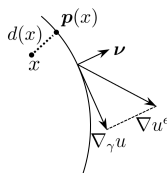
- surface FEMs (Dziuk 1998)
- trace FEMs (Olshanskii 2009)
- narrow band FEMs (Bertalmío 2001)

Surface Differential Operators

defined extrinsically

- $\gamma \subset \mathbb{R}^3$ is compact, **closed**, and orientable.
- $U_\delta = \{x \in \mathbb{R}^3 : \text{dist}(x, \gamma) < \delta\}$ is a tubular region.
- $d(x)$ is the **signed** distance function.
- $\nu(x) = \nabla d(x)$ is the unit outward normal vector.
- **Closest point projection**: $p(x) := x - d(x)\nu(x)$.
- **Tangential projection operator**: $\mathbf{P} := \mathbf{I} - \nu \otimes \nu$.
- **Extensions**: for v on γ , $v^e(x) := v \circ p(x) \quad \forall x \in U_\delta$.
- **Tangential gradient** and **surface divergence** of vector field g :
$$\nabla_\gamma v := \mathbf{P} \nabla v^e, \quad \text{div}_\gamma g := \text{tr}(\mathbf{P} \nabla g^e \mathbf{P}).$$
- **Laplace-Beltrami operator**:

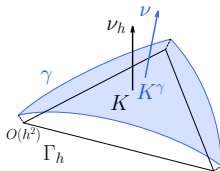
$$\Delta_\gamma := \text{div}_\gamma \nabla_\gamma.$$



Surface Finite Element Methods

a Galerkin method on the polyhedral surface approximated surface Γ_h

- Γ_h is a polyhedral surface.
- $\mathcal{T}_h$ is the set of faces of Γ_h .
- $h := \max_{K \in \mathcal{T}_h} \text{diam}(K)$.
- $K^\gamma := \mathbf{p}(K) \quad \forall K \in \mathcal{T}_h$.
- $\boldsymbol{\nu}_h$ is the outward unit normal (piecewise constant).



- $\mathbf{P}_h := \mathbf{I} - \boldsymbol{\nu}_h \otimes \boldsymbol{\nu}_h$.

- Approximation properties: $d(x) \lesssim h^2$, $|\boldsymbol{\nu} \circ \mathbf{p} - \boldsymbol{\nu}_h| \lesssim h$.

- Measures difference between γ and Γ_h :

$$d\sigma(\mathbf{p}(x)) = \mu_h(x) d\sigma_h(x), \text{ where } \mu_h \text{ satisfies } |1 - \mu_h| \lesssim h^2.$$

- Piecewise defined differential operators and Sobolev spaces on Γ_h :

$$\nabla_{\Gamma_h}, \text{div}_{\Gamma_h}, \Delta_{\Gamma_h}, \text{ and } H_h^m(\Gamma_h) := \{v \in L^2(\Gamma_h) : v|_K \in H^m(K), \forall K \in \mathcal{T}_h\}.$$

Surface Biharmonic Problem

- Surface biharmonic problem:

$$\Delta_\gamma^2 u = f \quad \text{on } \gamma, \quad \int_\gamma u \, d\sigma = 0.$$

where the source term satisfies $\int_\gamma f \, d\sigma = 0$.

Applying Green's formula on surface twice yields

$$(\Delta_\gamma^2 u, v)_\gamma = -(\nabla_\gamma \Delta_\gamma u, \nabla_\gamma v)_\gamma = (\Delta_\gamma u, \Delta_\gamma v)_\gamma.$$

- Variational formulation: Seek $u \in \dot{H}^2(\gamma)$ such that

$$(\Delta_\gamma u, \Delta_\gamma v)_\gamma = (f, v)_\gamma \quad \forall v \in \dot{H}^2(\gamma).$$

- Conformity:

$$u \in H^2(\gamma) \iff u \in H^1(\gamma), \nabla_\gamma u \in \mathbf{H}^1(\gamma) \iff u, \nabla_\gamma u \cdot \mathbf{n} \in H^1(\gamma).$$

For Γ_h , however, $\mathbf{H}^1(\Gamma_h)$ is generally not defined.

Surface Biharmonic Problem

- **Variational formulation:** Seek $u \in \dot{H}^2(\gamma)$ such that

$$(\Delta_\gamma u, \Delta_\gamma v)_\gamma = (f, v)_\gamma \quad \forall v \in \dot{H}^2(\gamma).$$

- **Relation with weak form involving Hessian:**

$$(\Delta_\gamma u, \Delta_\gamma v)_\gamma - (\nabla_\gamma^2 u, \nabla_\gamma^2 v)_\gamma = (K_G \nabla_\gamma u, \nabla_\gamma v)_\gamma$$

plane	$K_G = 0$, two forms are equal; Prefer form with ∇^2 for stability.
γ	$K_G \neq 0$. Not equivalent ; Inapplicable with ∇_γ^2 .
Γ_h	norm $ \cdot _{H_h^2(\Gamma_h)} \geq \Delta_{\Gamma_h} \cdot _{L^2(\Gamma_h)}$.

- **Nonconformity** introduced by Γ_h .
- Cannot use form with Hessian, weaker norm only; **Stability issues** arise.

$\implies$ stronger continuity
finite element space.

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New Zienkiewicz-Type Element

continuous element with weak $\mathbf{H}(\text{div})$ gradient

(M.Wang Z.Shi J.Xu 2007)

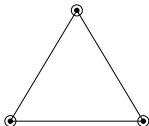
- Space function space:

$$V(K) := \mathcal{P}_2(K) + \text{span}\{q_{ij} : 1 \leq i < j \leq 3\}.$$
$$q_{ij} := \lambda_i^2 \lambda_j - \lambda_i \lambda_j^2 + \left(2(\lambda_i - \lambda_j) + \frac{3(\nabla \lambda_i - \nabla \lambda_j) \cdot \nabla \lambda_k}{|\nabla \lambda_k|^2} (2\lambda_k - 1) \right) \lambda_1 \lambda_2 \lambda_3 \quad k \neq i, j.$$

- Key property: For any $v \in V(K)$, $e \subset \partial K$, $\{a_1^e, a_2^e\} = \partial e$,

$$\frac{1}{|e|} \int_e \frac{\partial v}{\partial \mathbf{n}^e} ds = \frac{1}{2} \left(\frac{\partial v}{\partial \mathbf{n}^e}(a_1^e) + \frac{\partial v}{\partial \mathbf{n}^e}(a_2^e) \right).$$

- Degrees of freedom:



Key property $\Rightarrow \nabla v$ is weak $\mathbf{H}(\text{div})$,
 $v|_e \in \mathcal{P}_3(e) \Rightarrow v$ is continuous.

Question: How can we define vector-valued DoFs on Γ_h while preserving the properties?

Surface Piola Transforms

- Piola transform from $\mathcal{S}_0$ to $\mathcal{S}_1$: let $\Phi : \mathcal{S}_0 \rightarrow \mathcal{S}_1$ be a diffeomorphism, $D\Phi$ is the Jacobian of Φ . For a vector field $\mathbf{g}$ on $\mathcal{S}_0$:

$$(\mathcal{P}_\Phi \mathbf{g}) \circ \Phi := |D\Phi|^{-1} D\Phi \mathbf{g},$$

- tangential preservation: For surfaces $\mathcal{S}_0$ and $\mathcal{S}_1$ embedded in $\mathbb{R}^3$, if $\mathbf{g}$ is tangent to $\mathcal{S}_0$, then $\mathcal{P}_\Phi \mathbf{g}$ is tangent to $\mathcal{S}_1$.

- divergence preservation: let $\mu = |D\Phi|$ satisfies $\mu d\sigma_0 = d\sigma_1$,
$$\operatorname{div}_{\mathcal{S}_0} \mathbf{g} = \mu \operatorname{div}_{\mathcal{S}_1} (\mathcal{P}_\Phi \mathbf{g}), \quad \forall \mathbf{g} \in \mathbf{H}(\operatorname{div}_{\mathcal{S}_0}; \mathcal{S}_0).$$

- Piola transform from γ to Γ_h : $\check{\mathbf{g}} := \mathcal{P}_{\mathbf{p}^{-1}} \mathbf{g}$.
- Piola transform in vertex patch $\mathcal{T}_a$: let K_a be a fixed face, for $K \in \mathcal{T}_a$:

$$\mathcal{M}_a^K \mathbf{x} := \mathcal{P}_{\mathbf{p}_{K_a}^{-1}} \mathbf{x} = \left(\boldsymbol{\nu}_{K_a} \cdot \boldsymbol{\nu}_K \left[\mathbf{I} - \frac{\boldsymbol{\nu}_{K_a} \otimes \boldsymbol{\nu}_K}{\boldsymbol{\nu}_{K_a} \cdot \boldsymbol{\nu}_K} \right] \right) \mathbf{x}, \quad \forall \mathbf{x} \in \mathbb{R}^3.$$

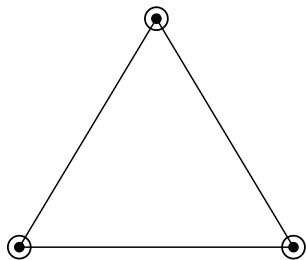
(A.Demlow M.Neilan 2023)

Surface NZT Element

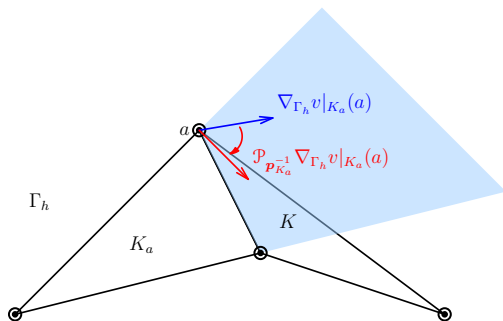
Definition

The **surface NZT finite element space** on the discretized surface Γ_h is defined as

$$V_h := \{v|_K \in V(K) : v|_K(a) = v|_{K_a}(a), \\ \nabla_{\Gamma_h} v|_K(a) = \mathcal{M}_a^K(\nabla_{\Gamma_h} v|_{K_a}(a)), \forall K \in \mathcal{T}_a, \forall a \in \mathcal{V}_h\}.$$

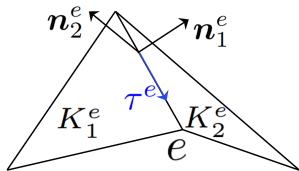


(d) DoFs of NZT element



(e) Connection of vertex gradient DoFs in V_h

Weak $\mathbf{H}(\operatorname{div}_{\Gamma_h}; \Gamma_h)$ Conformity



$$n_1^e \neq -n_2^e,$$

$$[\mathbf{g} \cdot \mathbf{n}] := \mathbf{g}|_{K_1^e} \cdot \mathbf{n}_1^e + \mathbf{g}|_{K_2^e} \cdot \mathbf{n}_2^e,$$

$$\mathbf{H}(\operatorname{div}_{\Gamma_h}; \Gamma_h) = \{\mathbf{g} \in \mathbf{H}_h^1(\Gamma_h) : [\mathbf{g} \cdot \mathbf{n}]_e = 0 \ \forall e \in \mathcal{E}_h\}.$$

Lemma (weak $\mathbf{H}(\operatorname{div}_{\Gamma_h}; \Gamma_h)$ conformity of $\nabla_{\Gamma_h} v$)

For any $v \in V_h$ and $e \in \mathcal{E}_h$, it holds that

$$\frac{1}{|e|} \int_e [\nabla_{\Gamma_h} v \cdot \mathbf{n}] ds_h = 0.$$

proof: By the Binet-Cauchy identity, for any $\mathbf{x} \in \mathbb{R}^3$, $[\mathcal{M}_a^K \mathbf{x} \cdot \mathbf{n}]_e = 0$

$\implies [\nabla_{\Gamma_h} v \cdot \mathbf{n}]_e(a) = 0$, similarly $[\nabla_{\Gamma_h} v \cdot \mathbf{n}]_e(b) = 0$.

+ key property of NZT space: $\frac{1}{|e|} \int_e \frac{\partial v}{\partial \mathbf{n}^e} ds = \frac{1}{2} \left(\frac{\partial v}{\partial \mathbf{n}^e}(a) + \frac{\partial v}{\partial \mathbf{n}^e}(b) \right)$.

H^1 -Conforming Relative

- $[\mathcal{M}_a^K \mathbf{x} \cdot \mathbf{n}]_e(a) = 0$ **but** $[\mathcal{M}_a^K \mathbf{x} \cdot \boldsymbol{\tau}]_e(a) \neq 0 \Rightarrow$ no longer continuous.
- H^1 -conforming relative: For any $v \in V_h$, define $\Pi_h^c v$ such that for any $K \in \mathcal{T}_h$, $a \in \partial e$, $e \subset \partial K$, it holds that $\Pi_h^c v|_K \in V(K)$ and

$$\Pi_h^c v(a) := v(a), \quad (\nabla_{\Gamma_h} \Pi_h^c v \cdot \boldsymbol{\tau}^e)(a) := \{\nabla_{\Gamma_h} v \cdot \boldsymbol{\tau}^e\}(a).$$

Lemma (H^1 -conforming relative)

For any $v \in V_h$, the H^1 -conforming relative $\Pi_h^c v$ satisfies

$$\|v - \Pi_h^c v\|_{L^2(K)} \lesssim h \|v\|_{L^2(K)}, \quad |v - \Pi_h^c v|_{H^1(K)} \lesssim h |v|_{H^1(K)} \quad \forall K \in \mathcal{T}_h.$$

- Applications:
 - discrete Poincaré inequality,
 - reduction of regularity assumptions for the exact solution.

Approximation Properties

- Interpolation to V_h : For $w \in C^1(\gamma) \cap H_h^3(\gamma)$,

$$\tilde{I}_h w(a) = w^e(a), \quad \nabla_{\Gamma_h} \tilde{I}_h w|_{K_a}(a) = \widetilde{\nabla_{\gamma} w}|_{K_a}(a), \quad \forall a \in \mathcal{V}_h.$$

- Interpolation to integral-free subspace $\mathring{V}_h$:

$$I_h w := \tilde{I}_h w - \frac{1}{|\Gamma_h|} \int_{\Gamma_h} \tilde{I}_h w \, d\sigma_h.$$

- Key estimate:

$$(\nabla_{\gamma} v)^e - \frac{1}{h} \left[\nabla_{\Gamma_h} v^e - \widetilde{\nabla_{\gamma} v}|_K - \mathcal{M}_a^K(\widetilde{\nabla_{\gamma} v}|_{K_a}) \right]$$

Lemma (approximation)

For $w \in C^1(\gamma) \cap H_h^3(\gamma) \cap \mathring{L}^2(\gamma)$, it holds that

$$\|w^e - I_h w\|_{H_h^m(\Gamma_h)} \lesssim h^{\min\{3-m, 2\}} \|w\|_{H_h^3(\gamma)} \quad 0 \leq m \leq 3,$$

$$\|\widetilde{\nabla_{\gamma} w} - \nabla_{\Gamma_h} I_h w\|_{H^m(K)} \lesssim h^{2-m} \|w\|_{H^3(K\gamma)} \quad \forall K \in \mathcal{T}_h, \quad 0 \leq m \leq 2.$$

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Stabilized Nonconforming FEM

- The bilinear form $a_h : V_h \times V_h \rightarrow \mathbb{R}$:

$$a_h(w, v) := (\Delta_{\Gamma_h} w, \Delta_{\Gamma_h} v)_{\Gamma_h} + \sum_{e \in \mathcal{E}_h} h_e^{-1} \int_e [\nabla_{\Gamma_h} w \cdot \mathbf{n}] [\nabla_{\Gamma_h} v \cdot \mathbf{n}] \, ds_h.$$

- The discrete linear functional $l_h : V_h \rightarrow \mathbb{R}$:

$$l_h(v) := \int_{\Gamma_h} f_h v \, d\sigma_h, \quad \text{where} \quad f_h := f^e - \frac{1}{|\Gamma_h|} \int_{\Gamma_h} f^e \, d\sigma_h.$$

- Stabilized nonconforming FEM: Find $u_h \in \mathring{V}_h$ such that

$$a_h(u_h, v_h) = l_h(v_h) \quad \forall v_h \in \mathring{V}_h.$$

No artificial parameters.

Stability

The discrete energy **semi**-norm:

$$\|v\|_h^2 := \|\Delta_{\Gamma_h} v\|_{L^2(\Gamma_h)}^2 + \sum_{e \in \mathcal{E}_h} h_e^{-1} \|[\nabla_{\Gamma_h} v \cdot \mathbf{n}]\|_{L^2(e)}^2 = a_h(v, v).$$

Lemma (discrete Poincaré inequality)

$$\|v\|_{L^2(\Gamma_h)} \lesssim |v|_{H_h^1(\Gamma_h)} \quad \forall v \in \mathring{V}_h.$$

proof: $\|\Pi_h^c v\|_{L^2(\Gamma_h)} \lesssim \|\Pi_h^c v\|_{H_h^1(\Gamma_h)} + \left| \int_{\Gamma_h} \Pi_h^c v d\sigma_h \right|.$

Theorem (energy norm)

$$\|v\|_{H_h^1(\Gamma_h)} \lesssim \|v\|_h \quad \forall v \in \mathring{V}_h.$$

proof: $|v|_{H_h^1(\Gamma_h)}^2 = -(\Delta_{\Gamma_h} v, v)_{\Gamma_h} + \langle [\nabla_{\Gamma_h} v \cdot \mathbf{n}] \{v\} \rangle_{\mathcal{E}_h} + \langle \{\nabla_{\Gamma_h} v \cdot \mathbf{n}\}, [v] \rangle_{\mathcal{E}_h}.$

Error Estimates

only utilizing $\|u\|_{H^3(\gamma)}$

Theorem (energy norm error estimate)

Let u be the solution of surface biharmonic problem on the C^4 surface γ , $u_h \in \mathring{V}_h$ be the numerical solution. Then,

$$\begin{aligned} \|(\Delta_\gamma u)^e - \Delta_{\Gamma_h} u_h\|_{L^2(\Gamma_h)} + \left(\sum_{e \in \mathcal{E}_h} h_e^{-1} \|[\nabla_{\Gamma_h} u_h \cdot \mathbf{n}]\|_{L^2(e)}^2 \right)^{1/2} \\ \lesssim h(\|u\|_{H^3(\gamma)} + \|f\|_{L^2(\gamma)}). \end{aligned}$$

Theorem (broken H^1 norm error estimate)

Let u be the solution of surface biharmonic problem on the C^4 surface γ , $u_h \in \mathring{V}_h$ be the numerical solution. It holds that

$$\|u^e - u_h\|_{H_h^1(\Gamma_h)} \lesssim h^2(\|u\|_{H^3(\gamma)} + \|f\|_{L^2(\gamma)}).$$

Reduction of regularity assumption via H^1 -conforming relative:

$$\begin{aligned}
 & \langle \nabla_\gamma(\Delta_\gamma u) \cdot \mathbf{n}^{e^\gamma}, [v_h] \rangle_{\mathcal{E}_h^\gamma} \\
 &= ((\Delta_\gamma)^2 u, v_h^\ell)_\gamma + (\nabla_\gamma(\Delta_\gamma u), \nabla_\gamma v_h^\ell)_\gamma \\
 &= (f, (v_h - \Pi_h^c v_h)^\ell)_\gamma + (\nabla_\gamma(\Delta_\gamma u), \nabla_\gamma(v_h - \Pi_h^c v_h)^\ell)_\gamma.
 \end{aligned}$$

Error decomposition: **approximation error**; **geometry error**;
nonconforming error for weak $\mathbf{H}(\text{div})$ gradient;
nonconforming error for discontinuous (absent in planar cases).

$$\begin{aligned}
 & a_h(I_h u, v_h) - (f_h, v_h)_{\Gamma_h} \\
 &= (\Delta_{\Gamma_h} u_I - (\Delta_\gamma u)^e, \Delta_{\Gamma_h} v_h)_{\Gamma_h} \\
 & \quad + ((\Delta_\gamma u)^e, \Delta_{\Gamma_h} v_h)_{\Gamma_h} + (\nabla_{\Gamma_h}(\Delta_\gamma u)^e, \nabla_{\Gamma_h} v_h)_{\Gamma_h} \\
 & \quad - (\nabla_{\Gamma_h}(\Delta_\gamma u)^e, \nabla_{\Gamma_h}(v_h - \Pi_h^c v_h))_{\Gamma_h} \\
 & \quad - (\nabla_{\Gamma_h}(\Delta_\gamma u)^e, \nabla_{\Gamma_h} \Pi_h^c v_h)_{\Gamma_h} + (\nabla_\gamma(\Delta_\gamma u), \nabla_\gamma(\Pi_h^c v_h)^\ell)_\gamma \\
 & \quad + (f, (\Pi_h^c v_h)^\ell - v_h^\ell)_\gamma + (f, v_h^\ell)_\gamma - (f_h, v_h)_{\Gamma_h} \\
 & \quad + \sum_{e \in \mathcal{E}_h} h_e^{-1} \langle [\nabla_{\Gamma_h} u_I \cdot \mathbf{n}], [\nabla_{\Gamma_h} v_h \cdot \mathbf{n}] \rangle_e.
 \end{aligned}$$

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Problem on the Unit Sphere

energy norm error $O(h)$, broken H^1 norm error $O(h^2)$

Surface: $r = 1$.

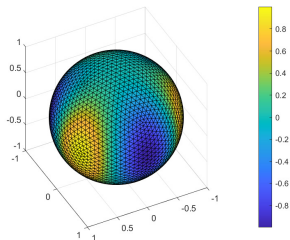
Exact solution: $u = r^{-3}(3x^2y - y^3)$.

$$E_0 = \|u^e - u_h\|_{L^2(\Gamma_h)},$$

$$E_1 = \|\nabla_{\Gamma_h} u^e - \nabla_{\Gamma_h} u_h\|_{L^2(\Gamma_h)},$$

$$E_\Delta = \|(\Delta_\gamma u)^e - \Delta_{\Gamma_h} u_h\|_{L^2(\Gamma_h)},$$

$$E_{\text{jump}} = \sum_{e \in \mathcal{E}_h} h_e^{-1} \|[\nabla_{\Gamma_h} u_h \cdot \mathbf{n}]\|_{L^2(e)}^2.$$



Dof	E_0	order	E_1	order	E_Δ	order	E_{jump}	order
486	7.54e-02		3.14e-01		2.11e-00		5.06e-01	
1926	1.91e-02	1.98	7.96e-02	1.98	1.03e-00	1.03	2.61e-02	0.96
7686	4.78e-03	2.00	1.99e-02	2.00	5.13e-01	1.01	1.31e-02	0.99
30726	1.19e-03	2.00	4.99e-03	2.00	2.56e-01	1.00	6.58e-02	1.00
122886	2.99e-04	2.00	1.25e-03	2.00	1.28e-01	1.00	3.29e-02	1.00

Table: Error table for surface NZT element approximation on the unit sphere.

Problem on a Torus

energy norm error $O(h)$, broken H^1 norm error $O(h^2)$

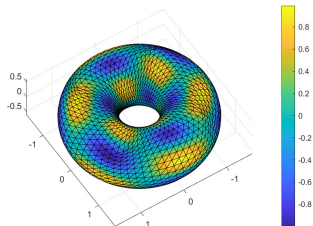
Surface: $R = 1, \quad r = 0.6$.

$$x = (R + r \cos(\theta)) \cos(\phi),$$

$$y = (R + r \cos(\theta)) \sin(\phi),$$

$$z = r \sin(\theta).$$

Exact solution: $u = \sin(3\phi) \cos(3\theta + \phi)$.



Dof	E_0	order	E_1	order	E_Δ	order	E_{jump}	order
1536	7.92e-01		4.25e-00		3.99e+01		8.99e-00	
6144	2.26e-01	1.81	1.49e-00	1.51	2.15e+01	0.89	6.88e-00	0.39
24576	6.88e-02	1.72	4.29e-01	1.80	1.13e+01	0.93	3.91e-00	0.81
98304	1.73e-02	1.99	1.09e-01	1.98	5.83e-00	0.95	2.02e-00	0.95
393216	4.24e-03	2.02	2.67e-02	2.03	2.96e-00	0.98	1.04e-00	0.96

Table: Error table for surface NZT element approximation on the torus.

Problem on an Implicitly Defined Surface

energy norm error $O(h)$, broken H^1 norm error $O(h^2)$

Surface: $(x - z^2)^2 + y^2 + z^2 - 1 = 0$.

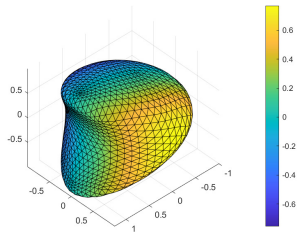
Exact solution: $u = y$.

$$E_0 = \|u^e - u_h\|_{L^2(\Gamma_h)},$$

$$E_1^* = \|\mathbf{P}_h(\nabla_\gamma u)^e - \nabla_{\Gamma_h} u_h\|_{L^2(\Gamma_h)},$$

$$E_\Delta = \|(\Delta_\gamma u)^e - \Delta_{\Gamma_h} u_h\|_{L^2(\Gamma_h)},$$

$$E_{\text{jump}} = \sum_{e \in \mathcal{E}_h} h_e^{-1} \|[\nabla_{\Gamma_h} u_h \cdot \mathbf{n}]\|_{L^2(e)}^2.$$



Dof	E_0	order	E_1^*	order	E_Δ	order	E_{jump}	order
3462	5.50e-01		8.17e-01		2.15e-00		1.33e-01	
13830	1.88e-01	1.55	2.81e-01	1.54	1.30e-00	0.72	8.04e-02	0.73
55302	5.20e-02	1.85	7.84e-02	1.84	6.99e-01	0.90	4.17e-02	0.95
221190	1.31e-02	1.99	1.98e-02	1.98	3.52e-01	0.99	2.06e-02	1.02
884742	3.27e-03	2.01	4.93e-03	2.01	1.76e-01	1.00	1.02e-02	1.02

Table: Error table for surface NZT element approximation on the implicitly defined surface.

THANK YOU!

Shuonan Wu and Hao Zhou. A stabilized nonconforming finite element method for the surface biharmonic problem, submitted, *arXiv preprint arXiv:2411.02952*.